Machine Learning Project Report

Weighted Gated Aggregation of Heterogeneous Deep Learning Models

for Privacy-Preserving Multi-Label 12-Lead ECG Classification

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Abstract

Automatic interpretation of the 12-lead electrocardiogram (ECG) can speed up the detection of myocardial infarction, ischaemic ST/T changes, conduction disturbances and hypertrophy. The deep networks that perform best, however, are trained on centrally pooled data, and hospitals usually cannot share patient records. Federated learning removes this constraint, but common methods such as FedAvg average model parameters over many communication rounds and therefore force every institution to train the same architecture.

This project proposes a one-shot federated ensemble in which four independent clients train structurally different networks (a 1D CNN, a 1D ResNet, a bidirectional LSTM and a CNN-LSTM) on disjoint, patient-separated shares of the data. Each client calibrates its model with temperature scaling and shares only class probabilities with a central gate, never data or parameters. The gate combines the probabilities with weights on the probability simplex; equal, learned static and per-patient dynamic weights are compared by patient-grouped cross-validation, and a more complex combiner is chosen only if it improves macro AUROC by at least 0.002.

Experiments use 21,388 records of the public PTB-XL dataset for multi-label classification of five diagnostic superclasses (NORM, MI, STTC, CD, HYP), with folds 1–8 for client training, fold 9 for validation and an untouched fold 10 for testing, over three random seeds. The equal-weight ensemble was selected and reaches a test macro AUROC of 0.905, a macro F1 of 70.7% and a label-wise accuracy of 86.9%. It beats the best single client in every seed (+0.009 to +0.012 AUROC, significant under a paired bootstrap) and comes within 0.004 AUROC of a model trained centrally on all the data and within 0.002 of FedAvg, which needs 20 rounds and a shared architecture. Integrated Gradients show which leads drive each prediction, and a per-finding triage step reports 53.7% of findings automatically at 0.937 precision while referring uncertain ones to a cardiologist. Most of the benefit of pooling ECG data is therefore obtained without exchanging data, parameters or architectures.

Keywords: *electrocardiogram, PTB-XL, federated learning, heterogeneous ensemble, gated aggregation, deep learning, explainable AI, clinical triage*